

ANALYSIS AND IMPROVEMENT OF A SEMI-LAGRANGIAN EXPONENTIAL SCHEME FOR THE SHALLOW-WATER EQUATIONS ON THE ROTATING SPHERE

JOÃO GUILHERME CALDAS STEINSTRASSER¹, MARTIN SCHREIBER^{2,3} AND
PEDRO S. PEIXOTO¹

Abstract. In this work, we study and extend a class of semi-Lagrangian exponential methods, which combine exponential time integration techniques, suitable for integrating stiff linear terms, with a semi-Lagrangian treatment of nonlinear advection terms. Partial differential equations involving both processes arise for instance in atmospheric circulation models. Through a truncation error analysis, we show that previously formulated semi-Lagrangian exponential schemes are limited to first-order accuracy due to the approximation of the integration factor acting on the discretization of the linear term; we then formulate a new discretization leading to second-order accuracy. Also, a detailed stability study is conducted to compare several Eulerian and semi-Lagrangian exponential schemes, as well as a well-established semi-Lagrangian semi-implicit method, which is used in operational atmospheric models. Numerical simulations of the shallow-water equations on the rotating sphere are performed to assess the orders of convergence, stability properties, and computational cost of each method. The proposed second-order semi-Lagrangian exponential method was shown to be more stable and accurate than the previously formulated schemes of the same class at the expense of larger wall-clock times; however, the method is more stable and has a similar cost compared to the well-established semi-Lagrangian semi-implicit method; therefore, it is a competitive candidate for potential operational applications in atmospheric circulation modeling.

2020 Mathematics Subject Classification. 65M12, 65M22, 76U60.

Keywords and phrases: Exponential integrator, semi-Lagrangian, accuracy and stability analysis, shallow-water equations on the rotating sphere, atmospheric modeling

¹ Applied Mathematics, Universidade de São Paulo, São Paulo, São Paulo, CEP 05508-090, Brazil

² Université Grenoble Alpes / Laboratoire Jean Kuntzmann, Saint-Martin-d'Hères, 38400, France; Inria AIRSEA team, Grenoble, 38058, France

³ Technical University of Munich, Garching, 85748, Germany

1. INTRODUCTION

Exponential integration methods are a large class of time-stepping schemes that have raised increasing interest for the integration of time-dependent partial differential equations containing stiff linear operators, since they allow integrating exactly (or at least very accurately) the linear terms of the governing equations, thus overcoming stability constraints related to their stiffness [16, 24]. These methods require the evaluation of matrix exponentials, which is a main challenge for their efficient implementation. The recent development of accurate and efficient linear algebra methods has directly contributed to the development and popularization of exponential integration schemes [24], which have been increasingly seen as viable alternatives to overcome Courant-Friedrichs-Lewy (CFL)-related time step limitations usually found in complex fluid dynamics problems such as in atmospheric modeling [13].

The development of exponential integration methods goes back to 1960, by [6], in which the variation-of-constants formula was used to derive the scheme, leading to the family of Exponential Time Differencing (ETD) methods. An alternative approach, proposed by [19], consists of defining a change of variables leading to a differential equation in which the linear operator appears only as arguments of exponential functions, a class of exponential schemes known as Integrating Factor (IF) methods. Both classes are derived considering a partition of the governing equations into linear and nonlinear terms. A third well-known class of exponential schemes is composed of the exponential Rosenbrock methods [15], which are derived from a linearization of the full time evolution operator, thus requiring the computation of its Jacobian. Since then, several ETD, IF and Rosenbrock exponential methods have been proposed, as well as novel classes of exponential schemes. Detailed reviews on exponential integration methods can be found, *e.g.*, in [16, 24]. These methods have been studied and applied in a very diverse range of contexts. In particular, several recent works have focused on geophysical fluid dynamics, in particular atmospheric circulation models, which is the application domain targeted in this work. [1, 7] assessed good stability properties and improved accuracy of Rosenbrock-type exponential schemes, compared to more traditional semi-implicit methods, when applied to the shallow-water equations (SWE) on the rotating sphere, but highlighted the need for more efficient approaches for computing the exponential of large matrices. Similar conclusions have been drawn by [2] comparing various time integration schemes in the context of the one-dimensional SWE. The challenge of improving the exponential computation has been tackled by [13, 21, 42]. Applications to more complex, three-dimensional atmospheric models have been investigated by [28, 29], with stable and accurate integration of both slow and fast atmospheric dynamics. Finally, methods considering rational approximations to the exponential functions, which lead to parallel-in-time methods, have been investigated for the SWE on the rotating sphere by [32, 33].

More recently, [26] proposed a class of semi-Lagrangian Runge-Kutta-type ETD schemes (SL-ETDRK) to tackle advective PDEs, inspired by the Eulerian ETDRK methods proposed by [8]. Semi-Lagrangian schemes, which follow the trajectories of particles along each time step instead of relying on fixed spatial grids as in Eulerian approaches, are popular methods in atmospheric modeling due to their improved stability properties. This improved stability was indeed verified for the proposed SL-ETDRK schemes in [26], through the numerical simulation of the SWE on the bi-periodic plane with a spectral discretization in space. A semi-Lagrangian exponential method has also been developed by [35], based on the Rosenbrock exponential approach, leading to a second-order scheme whose accuracy was confirmed on numerical tests of the SWE on the rotating sphere considering a cubed-sphere discretization.

This work proposes a better analytical understanding of convergence and stability properties of the SL-ETDRK schemes, as well as improvements for these methods. Indeed, although these methods have been derived analogously to their respective Eulerian counterparts ETD1RK and ETD2RK (respectively first- and second-order accurate) [8], the expected second-order convergence of SL-ETD2RK is not verified in practice. The numerical simulations presented by [26] reveal that both SL-ETD1RK and SL-ETD2RK are first-order accurate. Despite some indications that it could be related to the approximation for the integration factor arising on the derivation of the schemes, its influence on the overall accuracy of SL-ETD2RK was not completely clear. Here, we prove that the observed first-order accuracy of SL-ETD2RK is indeed related to the approximation of the integration factor, more specifically in the discretization of the linear term, which involves both an exponential operator and a spatial interpolation coming from the semi-Lagrangian approach. This issue is naturally absent in Eulerian schemes, in which the exponential integrates exactly the linear term. Then, we propose an alternative discretization, consisting of a splitting of the exponential operator, and prove that it leads indeed to second-order accuracy. We also conduct a detailed analytical stability study of both Eulerian and semi-Lagrangian ETDRK schemes, as well as the popular semi-Lagrangian semi-implicit SETTLS method [17] used in operational atmospheric models; we show in particular that standard linear stability analyses are not able to provide concluding insights on the stability of semi-Lagrangian exponential schemes due to a lack of commutation between exponential and interpolation operators; therefore, we conduct an empirical simulation-based stability study following a procedure described by [27]. Finally, we perform numerical experiments for the assessment of convergence and stability of the proposed improved schemes, which demonstrate its advantages compared to the previous Eulerian and semi-Lagrangian exponential methods and the semi-Lagrangian semi-implicit SETTLS.

In terms of application, our focus here is on the use of semi-Lagrangian exponential methods in the context of atmospheric modeling. Atmospheric circulation models are characterized by a vast range of waves propagating at different speeds, with the fast ones being responsible for often severe stability constraints, which has motivated the development of various discretization methods searching to establish a good trade-off between stability, accuracy, and computational cost [43]; therefore, combining the semi-Lagrangian and exponential approaches could better handle both advection processes and the stiffness of the linear terms. In particular, we expect semi-Lagrangian exponential methods to be more stable than Eulerian exponential ones, and more accurate than traditional semi-Lagrangian schemes (at least when stiff, linear processes have relatively important magnitude), such that good compromises between computational cost and accuracy could be obtained. We consider, for the numerical experiments and stability analyses conducted here, the SWE on the rotating sphere, using challenging benchmark test cases commonly used in atmospheric modeling research; thus, this work not only analyses and improves the semi-Lagrangian exponential methods proposed by [26], but it also extends their application domain since [26] focused on the SWE on the plane. Finally, we also compare the exponential semi-Lagrangian scheme to the well-established SL-SI-SETTLS, in order to assess the competitiveness of semi-Lagrangian exponential methods.

This paper is organized as follows: the shallow-water equations on the rotating sphere are introduced in Section 2; the time integrators considered in this work (the Eulerian ETDRK schemes by [8], the SL-ETDRK schemes by [26] and SL-SI-SETTLS by [17]) are presented in Section 3; a truncation error analysis indicating the sources of the first-order accuracy of the SL-ETD2RK scheme is developed in Section 4; an effectively second-order accurate semi-Lagrangian exponential method is proposed in Section 5, as well as its truncation error and computational complexity